



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MADESHWARA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

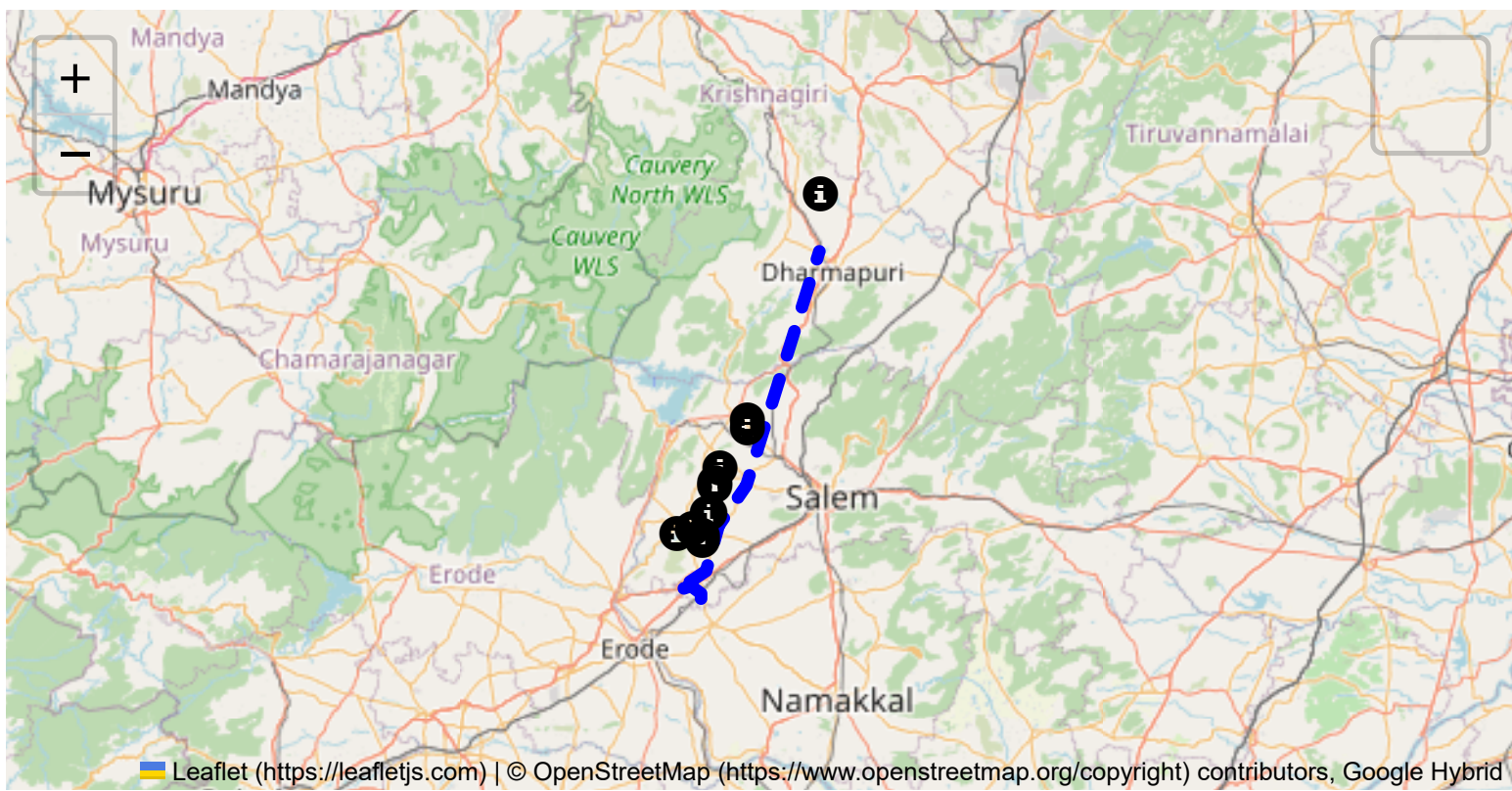
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

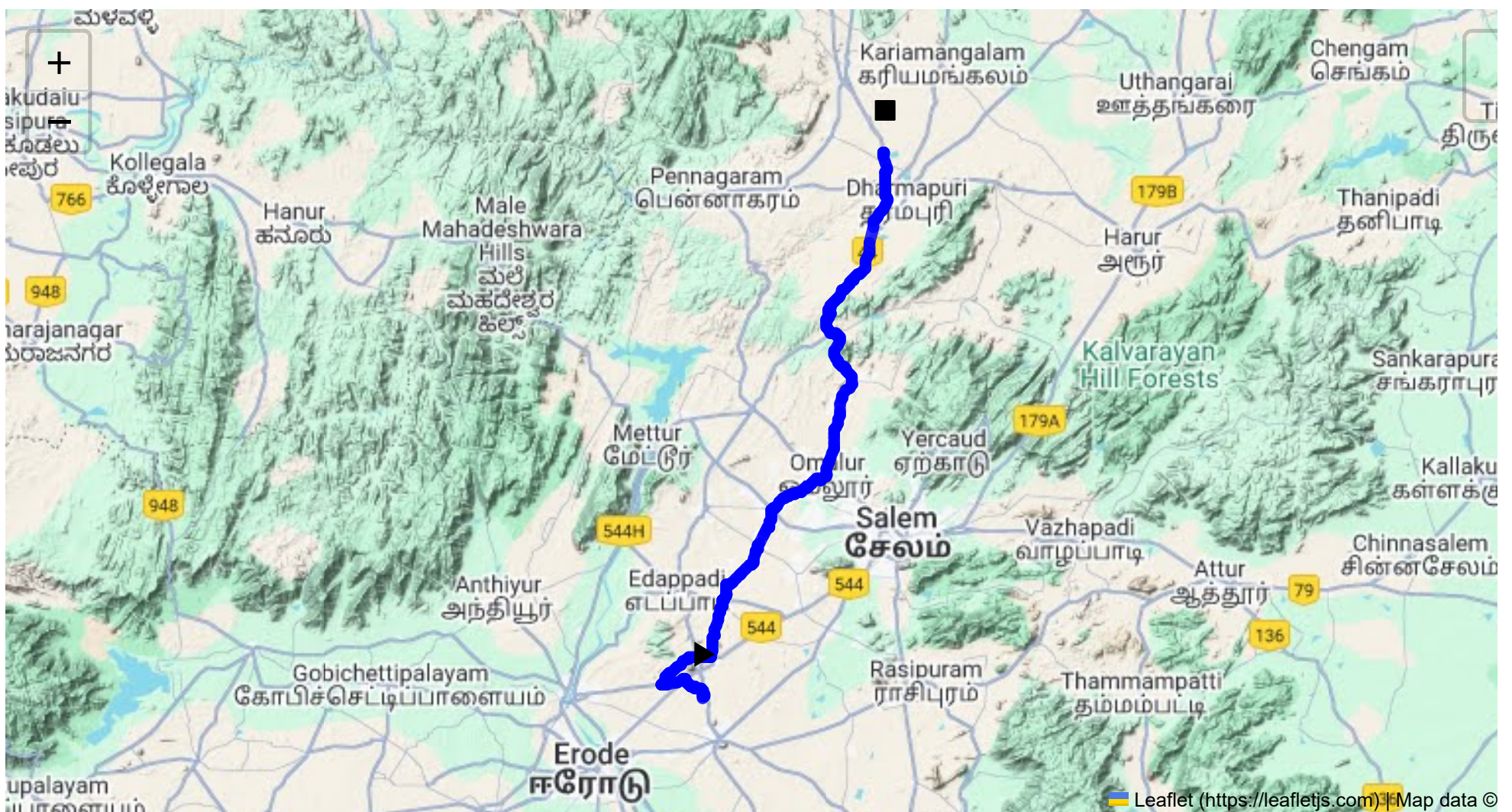
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 110.38 km
Estimated Duration: 2.4 hours
Adjusted Duration (Heavy Vehicle): 3.0 hours
Start: (11.4381, 77.8734)
End: (12.181293, 78.127879)



Welcome to the Journey Risk Management Study

Route Overview:

The route from CVQF+23W in Sangagiri to 54JH+G5 Mallikuttai spans approximately 110.38 km, primarily following state highways and main roads in Tamil Nadu. The journey typically takes around 2.39 hours for heavy vehicles.

1. Typical Weather Conditions:

The region generally experiences a tropical climate with hot summers, mild winters, and moderate to heavy monsoon rains. Potential weather-related hazards include:

- **Rainy Season (June to September):** Risk of flooding and reduced visibility.
- **Summer (March to June):** High temperatures could affect vehicle performance.

2. Traffic Patterns:

- **Peak Hours:** Typically occur in the morning (8 AM - 10 AM) and evening (5 PM - 8 PM).
- **Congestion-Prone Areas:**
 - Sankari RS - Erode Privu Rd.
 - Bhavani Main Road through Sankari West and Padaiveedu.

3. Road Quality and Infrastructure:

- Most roads are two-lane highways with varying road quality.
- Potholes and uneven surfaces may be prevalent, especially post-monsoon.
- Signage may be inadequate in rural stretches.

4. Alternative Routes:

In case of emergencies, major detours could include:

- Use of NH544 connecting Salem and Coimbatore for quicker access to bigger towns and infrastructure.

5. Local Regulations:

- Restrictions on hazardous material transportation may apply, particularly during nighttime or in certain populated regions.
- Specific permits may be required, enforced by local transport authorities.

6. Historical Incidents:

- While detailed incident records may not be publicly available, heavy vehicle accidents often occur due to overtaking, speeding, and road quality.

7. Environmental Considerations:

- Local wildlife and vegetation zones may require reduced speeds and caution.
- Areas near water bodies, especially during monsoons, may become sensitive to environmental protection rules.

8.Communication Coverage:

- Urban areas along the route have good mobile network coverage.
- Some rural stretches might experience coverage gaps; satellite-based communication tools are advisable for uninterrupted contact.

9.Emergency Response Times:

- Urban and suburban areas could have response times of around 15-30 minutes due to proximity to hospitals and fire stations.
- Rural and remote segments may face delays up to an hour or more.

10.Overall Risk Assessment:

- **Moderate Risk:** Due to traffic patterns, road quality, and moderate weather variance.
- **Key Risk Factors:** Weather conditions during rainy seasons, traffic congestion, and infrastructure variability.
- **Recommendations:** Enhanced driver training on route adaptability and safety protocols, utilizing technology for real-time weather and traffic updates, and ensuring compliance with local transport regulations.

It is critical for drivers transporting hazardous materials to adhere to safety regulations, maintain caution through variable road conditions, and remain informed about local traffic and weather updates.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45847, 77.85140	30 KM/Hr
5	Turn	Medium	11.45898, 77.85135	30 KM/Hr
6	Turn	High	11.46303, 77.84945	15 KM/Hr
7	Turn	Medium	11.45571, 77.81702	30 KM/Hr
8	Turn	High	11.45615, 77.81677	15 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
11	Turn	High	11.56096, 77.89931	15 KM/Hr
12	Turn	High	11.57035, 77.90223	15 KM/Hr
13	Turn	High	11.57041, 77.90221	15 KM/Hr
14	Turn	Medium	11.57649, 77.90610	30 KM/Hr
15	Turn	Medium	11.58279, 77.90658	30 KM/Hr
16	Turn	Medium	11.63833, 77.94818	30 KM/Hr
17	Turn	Medium	11.64171, 77.95184	30 KM/Hr
18	Turn	Medium	11.64243, 77.95198	30 KM/Hr
19	Turn	Medium	11.64721, 77.95118	30 KM/Hr
20	Turn	High	11.69595, 77.97114	15 KM/Hr
21	Turn	Medium	11.71386, 77.99552	30 KM/Hr
22	Turn	Medium	11.71735, 78.00019	30 KM/Hr
23	Turn	Medium	11.72765, 78.02291	30 KM/Hr
24	Turn	Medium	11.73786, 78.03505	30 KM/Hr
25	Turn	High	11.73652, 78.03866	15 KM/Hr
26	Turn	Medium	11.74110, 78.04575	30 KM/Hr
27	Turn	Medium	11.74257, 78.04708	30 KM/Hr
28	Turn	Medium	11.87022, 78.08380	30 KM/Hr
29	Turn	Medium	11.87913, 78.08206	30 KM/Hr
30	Turn	Medium	11.92566, 78.06503	30 KM/Hr
31	Turn	Medium	11.93699, 78.05412	30 KM/Hr
32	Turn	Medium	11.95810, 78.04971	30 KM/Hr
33	Turn	Medium	11.95865, 78.04991	30 KM/Hr
34	Turn	Medium	11.95963, 78.04954	30 KM/Hr
35	Turn	Medium	11.96072, 78.04963	30 KM/Hr
36	Turn	Medium	11.96281, 78.05124	30 KM/Hr
37	Turn	Medium	11.96384, 78.05141	30 KM/Hr
38	Turn	Medium	11.96442, 78.05104	30 KM/Hr
39	Turn	High	11.96557, 78.04930	15 KM/Hr
40	Turn	High	11.96657, 78.04930	15 KM/Hr
41	Turn	Medium	12.12041, 78.13284	30 KM/Hr
42	Turn	Medium	12.15023, 78.12980	30 KM/Hr
43	Turn	Medium	12.15624, 78.12990	30 KM/Hr
44	Turn	High	12.18519, 78.12678	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
45	Blind Spot	Blind Spot	12.18526, 78.12702	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	clinic	Sun Eye Care	11.6936242, 77.9730948	30 km/h	Medium
3	hospital	Dr. Pandiyarajan Hospital	11.692704, 77.969344	30 km/h	Medium
4	hospital	Dr. Sengodan Hospital	11.6959527, 77.9695325	30 km/h	Medium
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
7	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
8	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
9	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
10	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
12	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
14	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
15	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
16	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
17	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
18	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
19	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
20	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
21	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
22	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
23	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
24	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
25	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
26	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
28	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
30	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
31	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
32	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
13	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
27	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
29	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
33	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45898, 77.85135



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45571, 77.81702



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45615, 77.81677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57041, 77.90221



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58279, 77.90658



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69595, 77.97114



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71386, 77.99552



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71735, 78.00019



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72765, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73786, 78.03505



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73652, 78.03866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74110, 78.04575



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74257, 78.04708

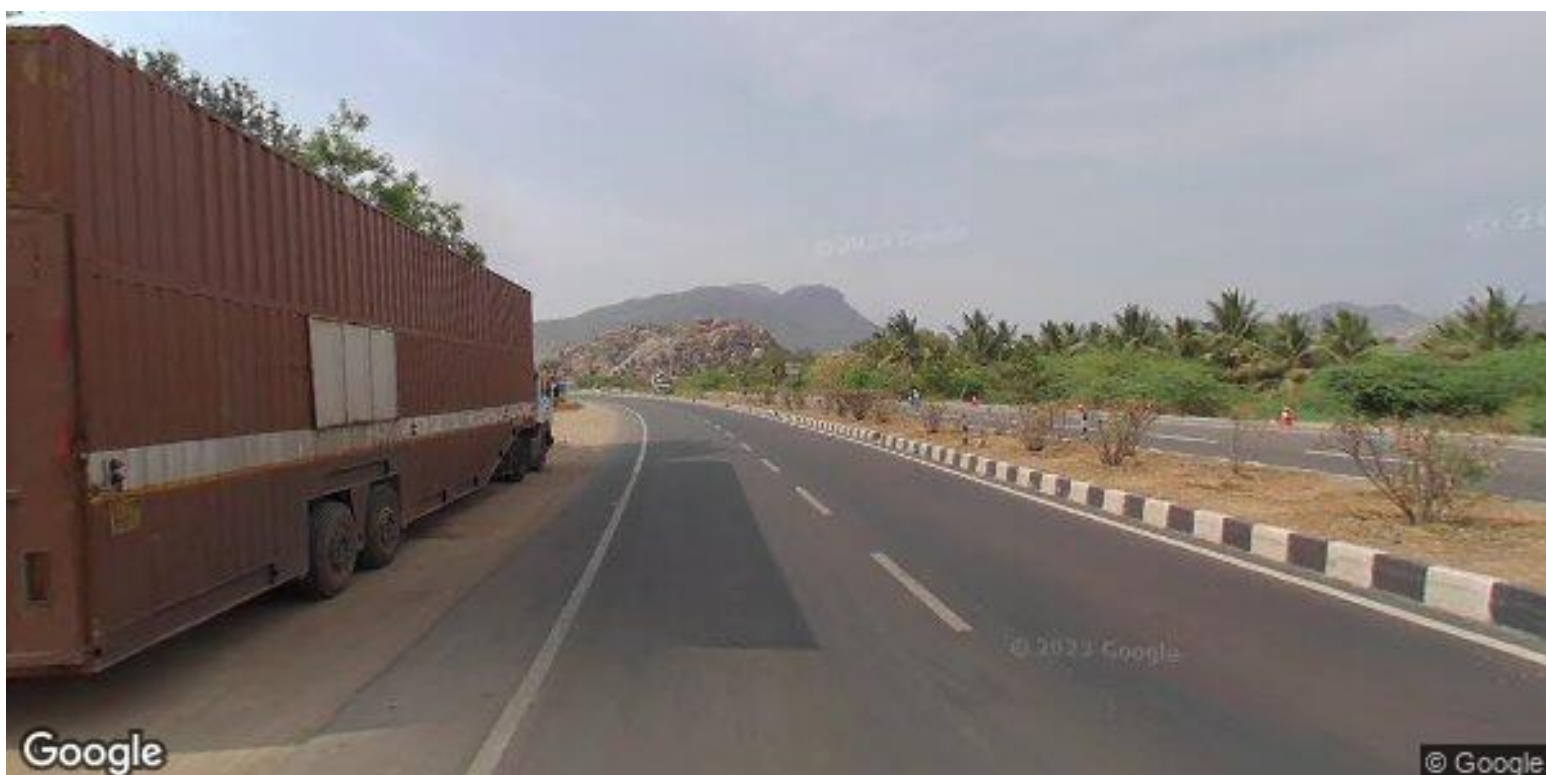


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87022, 78.08380



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87913, 78.08206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92566, 78.06503



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93699, 78.05412



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95810, 78.04971



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95865, 78.04991



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95963, 78.04954



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96072, 78.04963



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96281, 78.05124



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96384, 78.05141



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96442, 78.05104



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96557, 78.04930



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96657, 78.04930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12041, 78.13284



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.15023, 78.12980



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.15624, 78.12990



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.18519, 78.12678



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.18526, 78.12702
